

Creation Date 24-Nov-2010

Revision Date 29-Sep-2023

Revision Number 8

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

**Product Description:** Osmium tetroxide, 4 wt% solution in water  
**Cat No. :** 319010000; 319010050; 319010100  
**Synonyms** Osmic acid

**Unique Formula Identifier (UFI)** 1TM7-K3T4-CX0Q-9UG7

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### 1.3. Details of the supplier of the safety data sheet

#### Company

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

**E-mail address** begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

#### Poison Centre - Emergency information services

**Ireland** : National Poisons Information Centre (NPIC) -  
**01 809 2166** (8am-10pm, 7 days a week)  
**Malta** : +356 2395 2000  
**Cyprus** : +357 2240 5611

## SECTION 2: HAZARDS IDENTIFICATION

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## 2.1. Classification of the substance or mixture

### CLP Classification - Regulation (EC) No 1272/2008

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

Acute dermal toxicity

Category 2 (H310)

Acute Inhalation Toxicity - Vapors

Category 4 (H332)

Skin Corrosion/Irritation

Category 2 (H315)

Serious Eye Damage/Eye Irritation

Category 1 (H318)

Respiratory Sensitization

Category 1 (H334)

#### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Danger

### **Hazard Statements**

H310 - Fatal in contact with skin

H315 - Causes skin irritation

H318 - Causes serious eye damage

H332 - Harmful if inhaled

H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled

### **Precautionary Statements**

P310 - Immediately call a POISON CENTER or doctor/physician

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P332 + P313 - If skin irritation occurs: Get medical advice/attention

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P280 - Wear protective gloves/protective clothing/eye protection/face protection

## 2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors

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## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.2. Mixtures

| Component                                | CAS No     | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                      |
|------------------------------------------|------------|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------|
| Water                                    | 7732-18-5  | 231-791-2         | 96       | -                                                                                                                                      |
| Osmium oxide (OsO <sub>4</sub> ), (T-4)- | 20816-12-0 | EEC No. 244-058-7 | 4        | Acute Tox. 2 (H300)<br>Acute Tox. 1 (H310)<br>Acute Tox. 2 (H330)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318)<br>Resp. Sens. 1 (H334) |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                     |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

### 4.2. Most important symptoms and effects, both acute and delayed

Causes eye burns. May cause allergy or asthma symptoms or breathing difficulties if inhaled. Causes severe eye damage. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

### 4.3. Indication of any immediate medical attention and special treatment needed

|                           |                        |
|---------------------------|------------------------|
| <b>Notes to Physician</b> | Treat symptomatically. |
|---------------------------|------------------------|

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

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## Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

## Extinguishing media which must not be used for safety reasons

No information available.

## 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

## Hazardous Combustion Products

None under normal use conditions.

## 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

### 6.2. Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### 6.3. Methods and material for containment and cleaning up

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep container tightly closed. Keep refrigerated.

Technical Rules for Hazardous Substances (TRGS) 510  
Storage Class (LGK) (Germany)

Storage Class/LGK 6.1B

Switzerland - Storage of hazardous substances

Storage class - SC 6.1

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<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component                                | European Union | The United Kingdom                                                                                                           | France                                                                              | Belgium                                                                                                                                    | Spain                                                                                                                                                                                    |
|------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Osmium oxide (OsO <sub>4</sub> ), (T-4)- |                | STEL: 0.0006 ppm 15 min<br>STEL: 0.006 mg/m <sup>3</sup> 15 min<br>TWA: 0.0002 ppm 8 hr<br>TWA: 0.002 mg/m <sup>3</sup> 8 hr | TWA / VME: 0.0002 ppm (8 heures).<br>TWA / VME: 0.002 mg/m <sup>3</sup> (8 heures). | TWA: 0.0002 ppm 8 uren<br>TWA: 0.0016 mg/m <sup>3</sup> 8 uren<br>STEL: 0.0006 ppm 15 minuten<br>STEL: 0.0048 mg/m <sup>3</sup> 15 minuten | STEL / VLA-EC: 0.0006 ppm (15 minutos).<br>STEL / VLA-EC: 0.006 mg/m <sup>3</sup> (15 minutos).<br>TWA / VLA-ED: 0.0002 ppm (8 horas)<br>TWA / VLA-ED: 0.002 mg/m <sup>3</sup> (8 horas) |

| Component                                | Italy | Germany | Portugal                                               | The Netherlands | Finland                             |
|------------------------------------------|-------|---------|--------------------------------------------------------|-----------------|-------------------------------------|
| Osmium oxide (OsO <sub>4</sub> ), (T-4)- |       |         | STEL: 0.0006 ppm 15 minutos<br>TWA: 0.0002 ppm 8 horas |                 | STEL: 0.0002 ppm 15 minuutteina lho |

| Component                                | Austria                                                                                                                                                                                                                           | Denmark                                                                                                                                      | Switzerland                                                                                                                                    | Poland                                                                                | Norway                                                                                                                                                                           |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Osmium oxide (OsO <sub>4</sub> ), (T-4)- | Haut<br>MAK-KZGW: 0.0002 ppm 15 Minuten<br>MAK-KZGW: 0.002 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 0.0002 ppm 8 Stunden<br>MAK-TMW: 0.002 mg/m <sup>3</sup> 8 Stunden<br>Ceiling: 0.0002 ppm<br>Ceiling: 0.002 mg/m <sup>3</sup> | TWA: 0.0002 ppm 8 timer<br>TWA: 0.002 mg/m <sup>3</sup> 8 timer<br>STEL: 0.0004 ppm 15 minutter<br>STEL: 0.004 mg/m <sup>3</sup> 15 minutter | STEL: 0.0002 ppm 15 Minuten<br>STEL: 0.002 mg/m <sup>3</sup> 15 Minuten<br>TWA: 0.0002 ppm 8 Stunden<br>TWA: 0.002 mg/m <sup>3</sup> 8 Stunden | STEL: 0.006 mg/m <sup>3</sup> 15 minutach<br>TWA: 0.002 mg/m <sup>3</sup> 8 godzinach | TWA: 0.0002 ppm 8 timer<br>TWA: 0.002 mg/m <sup>3</sup> 8 timer<br>STEL: 0.0006 ppm 15 minutter. value calculated<br>STEL: 0.006 mg/m <sup>3</sup> 15 minutter. value calculated |

| Component                                | Bulgaria | Croatia                                                                                                                                                                          | Ireland                                                                                                                              | Cyprus | Czech Republic                                                                     |
|------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------------|
| Osmium oxide (OsO <sub>4</sub> ), (T-4)- |          | TWA-GVI: 0.0002 ppm 8 satima. Os<br>TWA-GVI: 0.002 mg/m <sup>3</sup> 8 satima. Os<br>STEL-KGVI: 0.0006 ppm 15 minutama. Os<br>STEL-KGVI: 0.006 mg/m <sup>3</sup> 15 minutama. Os | TWA: 0.002 mg/m <sup>3</sup> 8 hr. Os<br>TWA: 0.0002 ppm 8 hr. Os<br>STEL: 0.006 mg/m <sup>3</sup> 15 min<br>STEL: 0.0006 ppm 15 min |        | TWA: 0.002 mg/m <sup>3</sup> 8 hodinách. Os<br>Ceiling: 0.004 mg/m <sup>3</sup> Os |

| Component                                | Estonia | Gibraltar | Greece                                                                                               | Hungary                                                                                                                | Iceland                                                                                                  |
|------------------------------------------|---------|-----------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Osmium oxide (OsO <sub>4</sub> ), (T-4)- |         |           | STEL: 0.0006 ppm<br>STEL: 0.006 mg/m <sup>3</sup><br>TWA: 0.0002 ppm<br>TWA: 0.002 mg/m <sup>3</sup> | STEL: 0.004 mg/m <sup>3</sup> 15 percekben. CK Os<br>TWA: 0.002 mg/m <sup>3</sup> 8 órában. AK Os<br>lehetőséges borón | TWA: 0.0002 ppm 8 klukkustundum.<br>TWA: 0.002 mg/m <sup>3</sup> 8 klukkustundum.<br>Ceiling: 0.0004 ppm |

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|  |  |  |  |                        |                                  |
|--|--|--|--|------------------------|----------------------------------|
|  |  |  |  | keresztüli felszívódás | Ceiling: 0.004 mg/m <sup>3</sup> |
|--|--|--|--|------------------------|----------------------------------|

| Component                                | Latvia | Lithuania | Luxembourg | Malta | Romania                                                                            |
|------------------------------------------|--------|-----------|------------|-------|------------------------------------------------------------------------------------|
| Osmium oxide (OsO <sub>4</sub> ), (T-4)- |        |           |            |       | TWA: 0.001 mg/m <sup>3</sup> 8<br>or<br>STEL: 0.003 mg/m <sup>3</sup> 15<br>minute |

| Component                                | Russia | Slovak Republic                                  | Slovenia | Sweden | Turkey |
|------------------------------------------|--------|--------------------------------------------------|----------|--------|--------|
| Osmium oxide (OsO <sub>4</sub> ), (T-4)- |        | TWA: 0.0002 ppm<br>TWA: 0.0021 mg/m <sup>3</sup> |          |        |        |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

No information available

## Predicted No Effect Concentration (PNEC)

No information available.

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local

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conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

## Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

## Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls** No information available.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                          |                          |                                   |
|------------------------------------------|--------------------------|-----------------------------------|
| Physical State                           | Liquid                   |                                   |
| Appearance                               | Yellow                   |                                   |
| Odor                                     | Slight chlorine          |                                   |
| Odor Threshold                           | No data available        |                                   |
| Melting Point/Range                      | No data available        |                                   |
| Softening Point                          | No data available        |                                   |
| Boiling Point/Range                      | 100 °C / 212 °F          | @ 760 mmHg                        |
| Flammability (liquid)                    | No data available        |                                   |
| Flammability (solid,gas)                 | Not applicable           | Liquid                            |
| Explosion Limits                         | No data available        |                                   |
| Flash Point                              | No information available | Method - No information available |
| Autoignition Temperature                 | No data available        |                                   |
| Decomposition Temperature                | No data available        |                                   |
| pH                                       | 6 @ 20°C                 | 25 g/l aq.sol                     |
| Viscosity                                | No data available        |                                   |
| Water Solubility                         | Soluble                  |                                   |
| Solubility in other solvents             | No information available |                                   |
| Partition Coefficient (n-octanol/water)  |                          |                                   |
| Component                                | log Pow                  |                                   |
| Osmium oxide (OsO <sub>4</sub> ), (T-4)- | 0.9                      |                                   |
| Vapor Pressure                           | No data available        |                                   |
| Density / Specific Gravity               | 1.040                    |                                   |
| Bulk Density                             | Not applicable           | Liquid                            |
| Vapor Density                            | No data available        | (Air = 1.0)                       |
| Particle characteristics                 | Not applicable (liquid)  |                                   |

### 9.2. Other information

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

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## 10.2. Chemical stability

Stable under normal conditions.

## 10.3. Possibility of hazardous reactions

**Hazardous Polymerization**  
**Hazardous Reactions**

Hazardous polymerization does not occur.  
None under normal processing.

## 10.4. Conditions to avoid

Incompatible products.

## 10.5. Incompatible materials

Strong oxidizing agents. Halogens. Reducing Agent. Finely powdered metals.

## 10.6. Hazardous decomposition products

None under normal use conditions.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

##### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Category 2

Inhalation

Category 4

#### Toxicology data for the components

| Component                                | LD50 Oral        | LD50 Dermal | LC50 Inhalation           |
|------------------------------------------|------------------|-------------|---------------------------|
| Water                                    | -                | -           | -                         |
| Osmium oxide (OsO <sub>4</sub> ), (T-4)- | 15 mg/kg ( Rat ) | -           | LC50 = 40 ppm ( Rat ) 4 h |

##### (b) skin corrosion/irritation;

Category 2

##### (c) serious eye damage/irritation;

Category 1

##### (d) respiratory or skin sensitization;

Respiratory

Category 1

Skin

No data available

No information available

##### (e) germ cell mutagenicity;

No data available

##### (f) carcinogenicity;

No data available

There are no known carcinogenic chemicals in this product

##### (g) reproductive toxicity;

No data available

##### (h) STOT-single exposure;

No data available



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(i) STOT-repeated exposure; No data available  
Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

## 11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

Ecotoxicity effects Do not empty into drains.

### 12.2. Persistence and degradability

Persistence Not applicable for mixtures  
Soluble in water, Persistence is unlikely, based on information available.

### 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component                                | log Pow | Bioconcentration factor (BCF) |
|------------------------------------------|---------|-------------------------------|
| Osmium oxide (OsO <sub>4</sub> ), (T-4)- | 0.9     | No data available             |

### 12.4. Mobility in soil

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

### 12.5. Results of PBT and vPvB assessment

No data available for assessment.

### 12.6. Endocrine disrupting properties

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

### 12.7. Other adverse effects

Persistent Organic Pollutant This product does not contain any known or suspected substance  
Ozone Depletion Potential This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

Waste from Residues/Unused Products Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

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|                                       |                                                                                                                                                                                                                                                                                                            |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Contaminated Packaging</b>         | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                  |
| <b>European Waste Catalogue (EWC)</b> | According to the European Waste Catalog, Waste Codes are not product specific, but application specific.                                                                                                                                                                                                   |
| <b>Other Information</b>              | Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms.                                                                                              |
| <b>Switzerland - Waste Ordinance</b>  | Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600<br><a href="https://www.fedlex.admin.ch/eli/cc/2015/891/en">https://www.fedlex.admin.ch/eli/cc/2015/891/en</a> |

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

|                                                                 |                                                       |
|-----------------------------------------------------------------|-------------------------------------------------------|
| <b>14.1. UN number</b>                                          | UN3287                                                |
| <b>14.2. UN proper shipping name</b><br>Technical Shipping Name | Toxic liquid, inorganic, n.o.s.<br>(OSMIUM TETROXIDE) |
| <b>14.3. Transport hazard class(es)</b>                         | 6.1                                                   |
| <b>14.4. Packing group</b>                                      | II                                                    |

### ADR

|                                                                 |                                                       |
|-----------------------------------------------------------------|-------------------------------------------------------|
| <b>14.1. UN number</b>                                          | UN3287                                                |
| <b>14.2. UN proper shipping name</b><br>Technical Shipping Name | Toxic liquid, inorganic, n.o.s.<br>(OSMIUM TETROXIDE) |
| <b>14.3. Transport hazard class(es)</b>                         | 6.1                                                   |
| <b>14.4. Packing group</b>                                      | II                                                    |

### IATA

|                                                                 |                                                       |
|-----------------------------------------------------------------|-------------------------------------------------------|
| <b>14.1. UN number</b>                                          | UN3287                                                |
| <b>14.2. UN proper shipping name</b><br>Technical Shipping Name | Toxic liquid, inorganic, n.o.s.<br>(OSMIUM TETROXIDE) |
| <b>14.3. Transport hazard class(es)</b>                         | 6.1                                                   |
| <b>14.4. Packing group</b>                                      | II                                                    |

|                                                                      |                                  |
|----------------------------------------------------------------------|----------------------------------|
| <b>14.5. Environmental hazards</b>                                   | No hazards identified            |
| <b>14.6. Special precautions for user</b>                            | No special precautions required. |
| <b>14.7. Maritime transport in bulk according to IMO instruments</b> | Not applicable, packaged goods   |

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Water     | 7732-18-5 | 231-791-2 | -      | -   | X     | X    | KE-35400 | X    | -    |

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|------------------------------------------|------------|-----------|---|---|---|---|----------|---|---|
| Osmium oxide (OsO <sub>4</sub> ), (T-4)- | 20816-12-0 | 244-058-7 | - | - | X | X | KE-27435 | - | - |
|------------------------------------------|------------|-----------|---|---|---|---|----------|---|---|

| Component                                | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|------------------------------------------|------------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Water                                    | 7732-18-5  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Osmium oxide (OsO <sub>4</sub> ), (T-4)- | 20816-12-0 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Authorisation/Restrictions according to EU REACH

| Component                                | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------------------------------------|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Water                                    | 7732-18-5  | -                                                                   | -                                                                             | -                                                                                                     |
| Osmium oxide (OsO <sub>4</sub> ), (T-4)- | 20816-12-0 | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component                                | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|------------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Water                                    | 7732-18-5  | Not applicable                                                                            | Not applicable                                                                           |
| Osmium oxide (OsO <sub>4</sub> ), (T-4)- | 20816-12-0 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

### WGK Classification

Water endangering class = non-hazardous to waters (self classification)

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

# SAFETY DATA SHEET

Osmium tetroxide, 4 wt% solution in water

Revision Date 29-Sep-2023

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H310 - Fatal in contact with skin  
H332 - Harmful if inhaled  
H315 - Causes skin irritation  
H318 - Causes serious eye damage  
H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled  
H330 - Fatal if inhaled  
H300 - Fatal if swallowed  
H314 - Causes severe skin burns and eye damage

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Creation Date** 24-Nov-2010

**Revision Date** 29-Sep-2023

**Revision Summary** Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.**

# **SAFETY DATA SHEET**

Osmium tetroxide, 4 wt% solution in water

Revision Date 29-Sep-2023

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## **COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**